

Implementation of Peer-to-Peer Energy Trading via Solana Smart Contracts in a Permissioned Environment

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Thailand’s National Energy Trading Platform (NETP), envisaged since 2017 as a blockchain settlement layer for Peer-to-Peer (P2P) trading among rooftop-solar prosumers, remains in development on a Tendermint consortium chain, and the regulated buy-back gives prosumers no direct way to exchange surplus within the local distribution grid. This paper asks what a parallel-execution runtime carries in that role, measured on the three axes a settlement layer is bought on: throughput, latency, and compute cost. Six modular Anchor programs run on a permissioned Solana network with every high-frequency write routed to a per-entity program-derived account (PDA), so transactions with disjoint write sets execute in parallel under Sealevel, ordered by Proof-of-History (PoH); above them a continuous double auction (CDA) and a uniform-price auction clear the book over fifteen-minute epochs. Under that partition the per-meter write costs ≈ 13.5 k compute units (CU) and confirms at $p50 \approx 1.5\text{--}2.0$ s ($p95 \leq 3.2$ s), both unchanged from 80 to 200,000 meters; a simulated community-month (80 meters, 15% prosumers, 30 days at fifteen-minute intervals) replays at $1,858\times$ real time with zero delivery loss and fourteen chain-derived conservation assertions all closing. The finding is what binds once partitioning removes computation as the limit: throughput is capped by a shared write-locked fee payer and single-node consensus rather than by per-entity contention, and settlement is packet-bound, not compute-bound. A match leaves half the 200 k CU budget unused, yet the 1,232-byte packet admits only one: denser settlement needs signatures repackaged, not more compute — a constraint transferable to any signature-verifying settlement layer. The price rule, not the platform, decides participation.

1. Motivation and Design Rationale

Thailand’s state utilities have advanced a National Energy Trading Platform (NETP) since 2017 for P2P trading among rooftop-solar prosumers, on a Tendermint-based consortium blockchain [1]. This article presents a Solana-based architecture as an alternative execution layer, motivated by NETP’s own finding that Tendermint outperformed Ethereum and Hyperledger Fabric — which suggests further gains from Solana’s parallel-execution model. P2P market design and blockchain-based energy trading are well surveyed [2], [3] and demonstrated in pilots such as the Brooklyn Microgrid [4]; what such work reports is throughput and compute cost, presuming computation to be the axis along which a settlement layer scales. No study places a parallel-execution runtime in that role, which is the gap this paper addresses. The design under test partitions all hot-path state into per-entity PDAs; throughput, latency and compute cost are then measured together, so that what binds each can be named separately rather than attributed to computation by default.

2. On-Chain Settlement Architecture

The settlement layer is six Anchor programs [5] (anchor-lang and anchor-spl 1.0.0) — **energy-token, governance, oracle, registry, trading, and treasury** — written in Rust and compiled to SBF bytecode. Programs invoke one another through cross-program invocation (CPI), authority delegated to program-derived addresses (PDAs) that sign without any private key. Routing every high-frequency write to a per-entity PDA keeps write sets disjoint, so such transactions never contend for one account lock and execute in parallel under Sealevel. The permissioned environment is a private solana-test-validator network; all experiments ran on a single node (Apple M2, Agave 3.1.10). A complete run must satisfy three closure identities, each checkable from chain state alone:

$$\sum_{i,d} M(i,d) = E_{\text{settle}} = E_{\text{burn}}, \quad \sum_i R(i) = 10^{-3} \cdot \text{REC supply}, \quad \sum_m T_m = P_{\text{sell}} + W \quad (1)$$

Here $M(i,d)$ is the GRID minted for prosumer i on day d , E_{settle} and E_{burn} the energy settled and burned, $R(i)$ the REC certified to i , and T_m , P_{sell} , W the per-match buyer debit, seller proceeds, and wheeling charges. An audit harness re-derives the fourteen assertions behind Eq. (1) from chain state by RPC reads alone.

3. Simulation Setup and Dataset

The fleet uses the CINELDI 80-bus rural LV reference grid (SINTEF) [6] for topology only — the single-phase feeder with its V/A/W parameters — resolved for power flow with pandapower [7]. As CINELDI carries only hourly load and no generation, all time-series are simulator-produced: per-meter consumption and twelve prosumer nodes’ (15%) rooftop PV at fifteen-minute cadence, solar via pvlib [8] under a fixed seed (Table 1). Two further sweeps read per-write cost and latency against fleet size alone, firing the oracle write from N distinct meters over two epochs 61 s apart and together covering $N = 80$ to 200,000 (1.35 M transactions); the second ran unreset, its largest scale writing against 310,000 PDAs already on the ledger.

Table 1: The simulated community-month dataset (seed-deterministic; no field data).

Quantity	Value
fleet	80 meters (seeded solar/load physics); 12 prosumer sellers, 68 consumers
market policy	sell cap $C = 10$ kWh/day per prosumer; 0.1 kWh dust floor
horizon / readings	30 days $\times$ 96 ticks ($\Delta t = 15$ min) = 230,400 readings, integer Wh
month energy	15,868.5 kWh generated; 144,879.8 kWh consumed; 10,386.7 kWh interval surplus
oracle-accepted surplus	8,253.502 kWh; 919 readings gate-rejected

4. Results and Discussion

4.1. Market-Clearing Model

Both P2P rules clear one book. The uniform-price auction cumulates supply S ascending in ask and demand D descending in bid and takes the crossing pair of maximum volume, $(p^*, V^*) = \arg \max_{p_s \leq p_b} \min(S(p_s), D(p_b))$, priced at the marginal ask; the CDA fills each crossing pair pay-as-ask. Whichever rule sets price p on quantity q , one on-chain tariff taxes the seller in integer micros with floored division:

$$V = \lfloor qp/10^9 \rfloor, \quad f = \lfloor V\varphi/10^4 \rfloor, \quad W = \lfloor qw/10^9 \rfloor, \quad L = \lfloor V\lambda/10^4 \rfloor, \quad \text{net} = V - f - W - L \quad (2)$$

for fee $\varphi = 25$ bp and loss $\lambda = 5$ bp on value and wheeling $w = 1.15$ THB/kWh flat on energy. Per kWh the seller keeps $p(1 - \varphi - \lambda) - w = 0.997p - 1.15$ regardless of volume, breaking even against the 2.20 THB/kWh buy-back only above $p^\dagger \approx 3.36$.

4.2. Measured Outcomes and Limitations

Table 2 reports all three axes with what bounds each; throughput is client-observed confirmed goodput, with every non-confirmed transaction attributed to a failure class. **Compute is $O(1)$ and never binds:** the per-meter write holds 13.3–13.6 k CU across the whole 2,500× range — the largest fleet writing against 310,000 already-resident PDAs at the same cost — and a match spends about half the default budget. **Latency is set by block time, not by the partition:** confirmation holds $p50 \approx 1.5$ – 2.0 s and $p95 \leq 3.2$ s over that range, flat where a contended design would degrade. Being send→first-seen-confirmed it is dominated by single-node block time (≈ 400 – 600 ms) and quantised by the ± 1.5 s poll — an upper bound on the runtime’s own cost, not a measurement of it.

Throughput is bound by serialization, not contention: steady ingest holds 426–455 TPS from 10,000 to 200,000 meters — flat, not proportional, because one write-locked gateway fee payer signs every transaction on one validator; the PDAs are disjoint, so the ceiling lifts with payer pooling and more nodes, not a finer partition. **Settlement density is packet-bound:** one match fits per transaction. Its two 77-byte signed orders serialize to a 186-byte instruction-data pair and are carried again, with signature and key, in two ≈ 189 -byte Ed25519 verification instructions, so a second match overruns the 1,232-byte packet; Address Lookup Tables compress account keys, not instruction data. Settlement stays exactly auditable, Eq. (1) holding under all three price rules.

Table 2: Measured outcomes by axis; *sweep* figures come from the 80 → 200,000-meter ingest experiment, the rest from the community-month replay.

Axis	Measurement	What bounds it
compute	13.3–13.6 k CU/reading (<i>sweep</i>) 107 k CU/match ($\approx 54\%$ of the 200 k budget)	nothing — $O(1)$ in fleet size and state
latency (<i>sweep</i>)	$p50 \approx 1.5$ – 2.0 s, $p95 \leq 3.2$ s, flat over the same range	block time + ± 1.5 s poll, not the partition
throughput	426–455 TPS steady ingest (<i>sweep</i>) ≈ 213 readings·s ^{−1} and ≈ 53 TPS order goodput in replay	one write-locked fee payer, one node
settlement density	1 match/tx	1,232-byte packet, not compute
replay wall-clock	1,394.7 s for 230,400 readings (1,858× real time)	end-to-end lifecycle cost
delivery loss	0 unconfirmed; 919 readings gate-rejected	validation, not lost throughput
audit	14/14 chain-derived assertions	exactness of settlement

A single validator bounds execution and locking only, not consensus in a three-utility consortium. Sweep delivery loss stays $\leq 0.46\%$, exactly zero across the later 1.02 M first-attempt sends; the 0.48% residual is a deliberate anomaly-gate probe. Telemetry reproduced bit-identically across four runs (213–232 readings·s^{−1}); lifecycle figures come from one canonical run, since nullifier PDAs make identical replay impossible on a persistent ledger.

4.3. Where the Prosumer’s Revenue Goes

A secondary, economic result follows from the same runs: all three rules move identical energy over one cleared month (1,111.4 kWh), so Eq. (2) alone sets the ranking. Flat wheeling takes 1,278 THB from each P2P rule and nothing from the buy-back — a third of the uniform auction’s 3,834 THB gross — while the two proportional charges take about 0.3%. The uniform auction nets 2.290 THB/kWh, above $p^\dagger$ and beating the 2.200 buy-back; the CDA fills cheaper rungs and nets 2.065. The ranking reverses once supply is uncapped, so wheeling policy, not the ledger, is the lever.

5. Conclusion

Per-entity PDA partitioning carries a community-scale P2P market on a permissioned Solana network: the full token life-cycle replays at 1,858× real time with nothing lost, the ledger conserving energy and currency exactly. Across the three axes the partition holds where it governs and yields where it does not: compute is $O(1)$ over a 2,500× sweep and never binds, latency is flat over that range and set by block time, and throughput is capped by one write-locked fee payer on one node, not by per-entity contention. Only settlement density resists: the packet, not the compute budget, limits a transaction to one match, and that is where denser settlement must be won. A multi-validator consortium mapped to EGAT, MEA, and PEA is the natural next step, suited to UEC–ASEAN collaboration.

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